

a ? [Proposition 4](#) in the appendix shows that the change in the ATE from multiplying the outcome by a large factor a is approximately $\log(a)$ times the treatment effect on the extensive margin,¹¹

$$E_P[m(aY(1)) - m(aY(0))] = (P(Y(1) > 0) - P(Y(0) > 0)) \cdot \log(a) + o(\log(a)). \quad (2)$$

Thus, the ATE for $m(Y)$ will tend to be more sensitive to finite changes in scale the larger is the extensive margin treatment effect. This implies that sensitivity analyses that assess how treatment effect estimates for $m(Y)$ change under finite changes in the units of Y —or equivalently, under finite changes of c in $\log(c + Y)$ —are roughly equivalent to measuring the size of the extensive margin.

Remark 3 (Extension to continuous treatments). We focus on ATEs for binary treatments for expositional simplicity, although similar results apply with continuous treatments. In [Appendix B.2](#), we show that when d is a continuous treatment, any treatment effect contrast that averages $m(aY(d))$ across possible values of d (i.e. a parameter of the form $\int \omega(d)E[m(aY(d))]$) is arbitrarily sensitive to scaling when there is an extensive margin effect.

Remark 4 (Extension to OLS estimands). It is worth noting that the results in this section show that population ATEs for $m(Y)$ are sensitive to the units of Y . These results are about *estimands*, and thus any consistent *estimator* of the ATE for $m(Y)$ will be sensitive to scaling (at least asymptotically). Thus, our results apply to ordinary least squares (OLS) estimators when they have a causal interpretation, but also to non-linear estimators such as inverse-probability weighting or doubly-robust methods. Nevertheless, given the prominence of OLS in applied work, and the fact that OLS is sometimes used for non-causal estimands, in [Appendix B.3](#) we provide a result specifically on the scale-sensitivity of the population regression coefficient for a random variable of the form $m(Y)$ on an arbitrary random variable X . Our result shows that the coefficients on X will be arbitrarily sensitive to the scaling of Y when the coefficients of a regression of $\mathbb{1}[Y > 0]$ on X are non-zero. Thus, the OLS estimand using a logarithm-like function on the left-hand side will be sensitive to scaling even when it does not have a causal interpretation.

Remark 5 (Statistical significance). Equation (2) shows that $P(Y(1) > 0) - P(Y(0) > 0)$ is the dominant term in $\theta(a)$ for large a , which suggests that the t -statistic for an estimator of $\theta(a)$ will generally converge to that for the analogous estimator of the extensive margin effect, $P(Y(1) > 0) - P(Y(0) > 0)$. [Proposition 7](#) in the appendix formalizes this intuition when the treatment effects are estimated via OLS: As a is made large, the t -statistic for $\hat{\theta}(a)$ converges to that for the extensive margin estimate. In our empirical analysis of papers in the *American Economic Review* below, we find that indeed the t -statistics for estimates of the ATE using $\text{arcsinh}(Y)$ are typically close to those for the extensive margin effect.

Remark 6 (When most values are large). Researchers often have the intuition that if most of the values of the outcome are large, then ATEs for transformations like $\log(1 + Y)$ or $\text{arcsinh}(Y)$ will approximate elasticities, since $m(Y) \approx \log(Y)$ for most values of Y . Indeed, in an influential paper, [Bellemare and Wichman \(2020\)](#) recommend that researchers using the $\text{arcsinh}(Y)$ transformation

¹¹We say $f(a) = o(g(a))$ if $\lim_{a \rightarrow \infty} |f(a)/g(a)| = 0$. That is, as $a \rightarrow \infty$, $|f(a)|$ grows strictly slower than $|g(a)|$.

should transform the units of their outcome so that most of the non-zero values of Y are large. The results in this section suggest—perhaps somewhat counterintuitively—that if one rescales the outcome such that the non-zero values are all large, the behavior of the ATE will be driven nearly entirely by the effect of the treatment on the extensive margin and *not* by the distribution of the potential outcomes conditional on being positive. Moreover, the rescaling can be chosen to generate any magnitude for the ATE if the treatment affects the extensive margin.

Remark 7 (Zero extensive margin). [Proposition 1](#) applies to settings where treatment has a non-zero effect on average on the extensive margin. This raises the question of whether the use of log-like transformations is justified in the absence of an extensive margin treatment effect. Our [Proposition 3](#) below implies that the ATE for any log-like transformation will be sensitive to the units of the outcome for at least some distribution with *strictly* positive outcomes, but perhaps not arbitrarily so in the sense of [Proposition 1](#) (see [Section 3.1](#) for further discussion). Moreover, if one were confident that the extensive margin effect were exactly zero for all individuals, one could recover the ATE in logs for individuals with positive outcomes by simply dropping individuals with $Y = 0$. The use of log-like transformations is thus somewhat difficult to justify even in settings without an extensive margin.

2.3 Empirical illustrations from the *American Economic Review*

We illustrate the results in this section by evaluating the sensitivity to scaling of estimates using the $\text{arcsinh}(Y)$ transformation in recent papers in the *American Economic Review* (*AER*). In November 2022, we used Google Scholar to search for “inverse hyperbolic sine” among papers published in the *AER* since 2018. We searched for papers using $\text{arcsinh}(Y)$ rather than $\log(1 + Y)$ since the former are easier to find with a simple keyword search. Our search returned 17 papers that estimate treatment effects for an arcsinh -transformed outcome.¹² Of these, 10 explicitly interpret the results as percentage changes or elasticities, and 6 of the remaining 7 do not directly interpret the units. See [Appendix Table 1](#) for a list of the papers and relevant quotes. Of the 17 total papers using $\text{arcsinh}(Y)$, 10 had publicly available replication data that allowed us to replicate the original estimates and assess their sensitivity to scaling.¹³ For our replications, we focus on the first specification using $\text{arcsinh}(Y)$ presented in a table in the paper, which we view as a reasonable proxy for the paper’s main specification using $\text{arcsinh}(Y)$.¹⁴

We assess the sensitivity of these results by re-running exactly the same procedure as in the original paper, except replacing $\text{arcsinh}(Y)$ with $\text{arcsinh}(100 \cdot Y)$. Thus, for example, if the original paper estimated a treatment effect for the arcsinh of an outcome measured in dollars, we use the same procedure to re-estimate the treatment effect for the arcsinh of the outcome measured in cents.

¹²We consider papers with both binary and non-binary treatments, as our theoretical results extend easily to non-binary treatments; see [Remark 3](#). Seven of the 10 papers we replicated used a binary treatment.

¹³We include one paper where there was a slight discrepancy between our replication of the original result and the result reported in the paper that only affected the third decimal place.

¹⁴We use the first coefficient presented in a figure for one paper without any tables in the main text using $\text{arcsinh}(Y)$. If the first specification is a validation check (e.g. a pre-trends test), we use the first specification of causal interest.

Since (2) shows that the sensitivity to scaling depends on the size of the extensive margin effect, we also estimate the extensive margin effect by using the same procedure as in the original paper but with the outcome $\mathbb{1}[Y > 0]$.

Paper	Treatment Effect Using:			Change from rescaling units:	
	$\text{arcsinh}(Y)$	$\text{arcsinh}(100 \cdot Y)$	Ext. Margin	Raw	%
Azoulay et al (2019)	0.003	0.017	0.003	0.014	464
Fetzer et al (2021)	-0.177	-0.451	-0.059	-0.273	154
Johnson (2020)	-0.179	-0.448	-0.057	-0.269	150
Carranza et al (2022)	0.201	0.453	0.055	0.252	125
Cao and Chen (2022)	0.038	0.082	0.010	0.044	117
Rogall (2021)	1.248	2.150	0.195	0.902	72
Moretti (2021)	0.054	0.068	0.000	0.013	24
Berkouwer and Dean (2022)	-0.498	-0.478	0.010	0.020	-4
Arora et al (2021)	0.113	0.110	-0.001	-0.003	-3
Hjort and Poulsen (2019)	0.354	0.354	0.000	0.000	0

Table 1: Change in estimated treatment effects from re-scaling the outcome by a factor of 100 in papers published in the *AER* using $\text{arcsinh}(Y)$

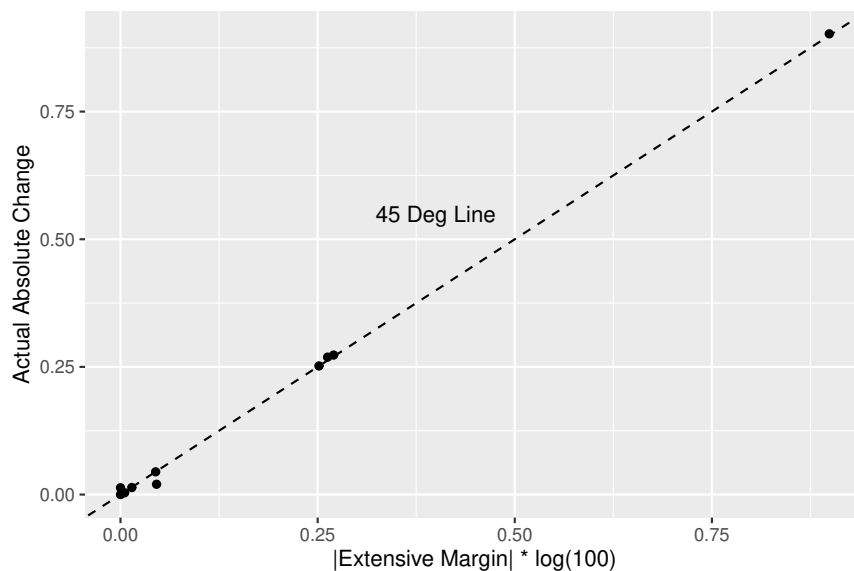
Note: This table replicates treatment effect estimates using $\text{arcsinh}(Y)$ as the outcome in recent papers published in the *AER*, and explores their sensitivity to the units of Y . The first column shows the author(s) and date of the paper. The second column shows the treatment effect on $\text{arcsinh}(Y)$ using the units originally reported in the paper. The third column shows a treatment effect estimate constructed identically to the estimate in column 2 except using $\text{arcsinh}(100 \cdot Y)$ as the outcome instead of $\text{arcsinh}(Y)$, e.g. converting Y from dollars to cents before taking the arcsinh transformation. The fourth column shows an estimate of the size of the extensive margin, obtained using $\mathbb{1}[Y > 0]$ as the outcome. The final two columns show the raw difference and percentage difference between the second and third columns. The table is sorted on the magnitude of the percentage difference.

The results of this exercise, shown in Table 1, illustrate that treatment effect estimates can be quite sensitive to the scaling of the outcome when the extensive margin is not approximately zero. Indeed, in 5 of the 10 replicable papers, multiplying the outcome by a factor of 100 changes the estimated treatment effect by more than 100% of the original estimate. The change in the estimated treatment effect is less than 10% only in three papers, all of which have either zero or near-zero (<1 p.p.) effects on the extensive margin. Figure 1 shows that the (absolute) change in the estimated treatment effect is larger when the extensive margin effect is larger, with the change lining up very closely with the approximation given in (2).¹⁵

Using the same 10 papers, we also estimate treatment effects using $\log(1 + Y)$ as the outcome, and analogously explore how the results change when we multiply the units of Y by 100. (Four of the 10 papers that we replicate report an alternative specification using $\log(1 + Y)$ in the paper.) The results, shown in Appendix Table 2, are qualitatively quite similar those in Table 1, with five

¹⁵In Appendix Figure 1, we plot the t -statistics for the treatment effects estimates as well as those for the extensive margin effect. In line with the discussion in Remark 5, we find that the t -statistics for the treatment effect using $\text{arcsinh}(Y)$ tend to be similar to those for the extensive margin, except when the extensive margin is very small, and become even closer when multiplying the units by 100.

Figure 1: Change from multiplying outcome by 100 versus extensive margin effect



Note: This figure shows the relationship between the sensitivity of treatment effects using $\text{arcsinh}(Y)$ to re-scaling the units of Y and the size of the extensive margin. For each replicated paper, this figure plots the absolute value of the change in the estimated treatment effect from multiplying the outcome by 100 (i.e. the absolute value of the Raw Change column in Table 1) on the y -axis against $\log(100)$ times the absolute value of the extensive margin effect on the x -axis. If the approximation in (2) were exact, all points would lie on the 45 degree line.

of the 10 treatment effect estimates again changing by more than 100%. These results underscore the fact that Proposition 1 applies to *all* log-like transformations, including both $\text{arcsinh}(Y)$ and $\log(c + Y)$ for any constant c .

3 Sensitivity to scaling for other ATEs

Our results so far show that ATEs for transformations that are defined at zero and approximate $\log(y)$ are arbitrarily sensitive to scaling. What other options are available when there are zero-valued outcomes? To help delineate alternative options, in this section we provide a result showing what properties a parameter defined with zero-valued outcomes can have. Specifically, we establish a “trilemma”: When there are zero-valued outcomes, there is no parameter that is (a) an average of individual-level treatment effects of the form $\theta_g = E_P[g(Y(1), Y(0))]$, (b) scale-invariant, and (c) point-identified.¹⁶ Any approach for settings with zero-valued outcomes must therefore abandon one of the properties (a)–(c); in Section 4 below we discuss several approaches that relax one (or more) of these requirements.

Before stating our formal result, we must make precise what we mean by scale-invariance and

¹⁶Of course, not all parameters of the form $E_P[g(Y(1), Y(0))]$ can be interpreted as an average of individual treatment effects. For example $E[\mathbb{1}[Y(1) > 0, Y(0) > 0]]$ is the fraction of individuals whose outcomes is positive under both treatments, rather than a treatment effect. Our results apply to all parameters of this form, regardless of whether they are average treatment effects *per se*.